Computers and Education: Artificial Intelligence 8 (2025) 100394

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

What percentage of secondary school students do their homework with the help of artificial intelligence? - A survey of attitudes towards artificial intelligence

Mátyás Turós *, Róbert Nagy, Zoltán Szűts

Eszterházy Károly Catholic University, Eszterházy tér 1, H-3300, Eger, Hungary

ARTICLE INFO

Keywords:
Education
Authorship
Student
Attitudes
Sentiment analysis
Educational technology

ABSTRACT

Based on practical experiences, news articles, and subjective reports, a significant proportion of high school students complete their writing assignments (essays) using ChatGPT or other text-generating software. However, to the best of our knowledge, there is a very limited amount of work on this topic. Therefore, our study has two objectives. First, through human coding, we determine whether student essays submitted as homework were created by humans or software, and second, we examine these essays' sentiment and authorship, as well as factors determining the use of artificial intelligence in learning. Data collection was conducted in April and May 2024 across thirty classes from twenty Hungarian high schools. Our results show that 16 % of high school students ($N = 548$) wrote their text-based homework assignments using artificial intelligence, though neither age nor parents' average educational level determined the use of AI in learning. Although women have more positive attitudes toward using AI for learning purposes, they are less likely to use it to complete their homework assignments. Finally, we found that fears related to artificial intelligence are general and lack specificity. Our findings have three practical implications: 1) they highlight the need for targeted educational programs to develop ethical AI use, 2) they assist teachers in identifying AI-generated texts, and 3) they draw attention to gender differences in AI usage.

1. Introduction

A vigorous societal and academic discourse is taking place regarding the impact of artificial intelligence on public education, higher education and academic life (Dergaa et al., 2023). This discourse has expanded beyond the field of data and information science in recent years, with numerous disciplines now examining the phenomenon within their own interpretative frameworks (Nguyen et al., 2023). Educational research naturally focuses primarily on the pedagogical context, specifically the integration of AI-based technologies into the learning process. Our study has dual objectives. Firstly, as research on students' use and perception of artificial intelligence requires empirical evidence (Casal-Otero et al., 2023), we focus on students' attitudes towards artificial intelligence and the factors determining its use. This objective is justified by assertions that knowledge related to artificial intelligence use will become as essential as reading and writing skills in the near future (Wojciechowski & Korjonen-Kuusipuro, 2023). Secondly, as the detection of artificial intelligence use in examination situations (Chiu, 2023) and misuse in homework assignments pose increasingly significant challenges for educators, we also investigate what proportion of secondary school students use software to complete their assigned essays, and which factors influence this practice. This objective is warranted by findings suggesting that whilst AI-solved tasks may impair knowledge acquisition processes and diminish creative and critical thinking abilities (Kooli, 2023), one study reported that 33 % of primary school students acknowledged using text-generation software for homework completion (Li et al., 2024). Due to the practical implications of artificial intelligence use in education, our study focuses on two areas amongst secondary school students: students' attitudes towards artificial intelligence, and the authorship of homework essays (whether human-authored or software-generated texts). We determine authorship not through student self-reporting or software detection, but through human coding. Our research questions are the following. Q1: What percentage of students use artificial intelligence to complete their writing assignments (essays)? Q2: What determines the sentiment of the essays? Q3: What determines essay authorship (human or software)?

* Corresponding author.
E-mail address: turos.matyas@uni-eszterhazy.hu (M. Turós).

https://doi.org/10.1016/j.caeai.2025.100394
Received 23 August 2024; Received in revised form 3 February 2025; Accepted 14 March 2025
Available online 15 March 2025
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Q4: What determines the use of artificial intelligence in learning?

1.1. Statement of contribution

Our research aims to contribute to the scientific discourse on the educational applications of artificial intelligence from multiple perspectives. Bibliometric analyses in the field indicate that whilst research in this area is intensive – particularly in Asia, Europe and North America – the majority of studies focus on higher education, whilst primary and secondary school applications remain under-researched (Fu et al., 2024; Tsekhmister et al., 2024). García-López et al. (2025) also emphasise the necessity of examining the technology in non-higher education settings. It is of particular significance that no similar research has been conducted in Hungary thus far. A further important consideration is that whilst earlier research focused on the technical aspects of artificial intelligence, in recent years attention has shifted to how artificial intelligence can directly support student learning and development (Yun et al., 2025). Ritter et al. (Ritter et al., 2024) and Giannakos et al. (2024) point out that GenAI cannot replace, but can only complement traditional educational methods, and finding the appropriate balance between artificial intelligence-based and traditional methods is crucial. Therefore, our study also examines whether the extent of artificial intelligence use is indeed a critical point, as the literature suggests that the best results are generally observed with moderate usage (Adewale et al., 2024).

Our research is also motivated by the fact that the educational application of artificial intelligence raises more ethical than logical questions (Illera, 2024). Successful implementation requires addressing ethical issues, establishing clear guidelines and introducing appropriate monitoring mechanisms (Hsu & Silalahi, 2024). Jose and Jose (2023) emphasise the need to establish appropriate digital monitoring systems to prevent cheating and plagiarism through artificial intelligence platforms. Research by Murtiningsih et al. (2024) also confirms that excessive use of ChatGPT reduces critical thinking and learner autonomy, whilst increasing the risk of cheating and plagiarism. The authors therefore recommend further research into data protection, ethical and equity considerations. Rojas and Chiappe (2024), and Ifenthaler et al. (2024) advocate for more thorough analysis of the ethical aspects of artificial intelligence in various educational contexts. Ali, Murray, et al. (2024), in their study examining the benefits and limitations of ChatGPT's educational applications, highlighted issues of plagiarism and authenticity, suggesting the development of critical thinking for evaluating artificial intelligence-generated content.

The attitudinal component of our research was inspired by the recommendation of Tsekhmister et al. (2024), who advocate exploring the relationships between student attitudes and artificial intelligence use. The construction of our hierarchical regression model was guided by theoretical considerations, although the application of similar models is not without precedent in the literature (cf. Bakhadirov et al., 2024; Cabero-Almenara et al., 2024).

2. Literature review

2.1. Issues in the use of artificial intelligence in learning

Artificial intelligence affects all the essential features of the teaching and learning process, influencing students' learning styles, changing the role of the teacher, the availability and constructability of information. Its use in education also poses risks and offers many opportunities (O. Ali, Murray, et al., 2024). For example, in the online space, often used by learners as a primary source of learning, where content may be unstructured and unregulated, AI blurs the boundaries between human and machine information, with potentially negative consequences. However, individualised tests based on personal performance data and adaptive learning pathways containing curriculum elements tailored to students' knowledge levels significantly contribute to maintaining student interest and motivation (Gligorea et al., 2023). According to Nguyen et al. (2023), analysis of student data can be used to predict students' future performance, making the learning process more personalised, data mining helps to understand the learning environment and students' behaviour, and personalised learning allows students to choose subjects that match their interests. The use of artificial intelligence in learning also supports equitable education systems that take into account learning disadvantages, different socio-cultural backgrounds and differentiation (Kochmar et al., 2022). Expectations and positive outcomes associated with AI (Fütterer et al., 2023; Rejeb et al., 2024) are also related to assessment and school management (Martínez-Comesaña et al., 2023) or psychological factors such as students' self-confidence (Chiu et al., 2023).

However, effectiveness - see the introductory chapter of this paper (Kooli, 2023) - does not seem to be a given in itself, i.e. not only with the availability of technology. Teachers and ethical use also play a crucial role in the correct design of AI-based teaching practices (Abulibdeh et al., 2024; Linderoth et al., 2024; Smith & Vickers, 2024). This view is supported by empirical results showing both advantages and disadvantages (Hadi Mogavi et al., 2024; Marzuki et al., 2023) and in a systematic review by Wang et al. (2024). In the latter case, while the authors acknowledge the potential benefits of AI in terms of accessibility, personalisation, automation and interactivity, they also point to response errors, dependency, a growing digital divide and privacy risks (Wang et al., 2024). Individual pedagogical work and organised secondary school and university programmes to develop AI literacy are therefore increasingly needed (Zeb et al., 2024), as they can help students learn about the potential harms of using AI in learning, methods to protect against these harms and ethical use (Choi et al., 2024).

Attitudes play a crucial role in shaping perceptions and practices related to AI (Ofosu-Ampong, 2024; Zhang et al., 2022). Thus, in order to realise the benefits of AI in education, it is not enough to provide training programmes, but it is also necessary to gain a better understanding of learners' attitudes towards AI, the factors that determine attitudes, and the practices related to its use. This assertion is supported by the literature (Kim & Lee, 2024; Sanusi et al., 2024; Stein et al., 2024), and our research questions Q2 and Q4 were guided by these considerations. However, before presenting our empirical study, we also review the methodological literature relevant to answering questions Q2 and Q4, which are related to our research questions Q1 and Q3.

2.2. Methodological issues in determining authorship (human or machine)

Monje et al. (2024) demonstrated that AI-generated texts contain fewer characters overall than human-generated texts (mean 3066 vs 7426; $p < .0001$), have a higher mean word length (mean 5.3 vs 4.8; $p < .0001$), and achieve lower Flesch-Kincaid scores (mean 46 vs 59; $p < .0001$). Using their proprietary detection method, the authors distinguished between AI-generated and human-generated text with 94.1 % sensitivity and 100 % specificity (AUC .99). Shah et al. (2023) examined syllable count, word length, sentence structure, functional word usage and punctuation ratio to detect AI-generated text, and subsequently applied machine learning models—including logistic regression, decision tree, random forest, support vector machine (SVM) classification and gradient boosting—which, in combination, classified AI- and human-generated text with 93 % accuracy.

Reviriego et al. (2023) utilised length-insensitive Maas and RTTR metrics to measure lexical diversity to determine the difference between human- and machine-generated texts, and established that ChatGPT-generated texts exhibited lower lexical diversity than human texts. A similar study was conducted by André et al. (2023), who examined differences between human and AI-generated texts through software comparison of linguistic and stylometric features. In their study, the authors analysed perplexity, grammar and N-gram distribution; the results revealed that human-generated texts possessed higher perplexity, contained more grammatical errors and demonstrated more

varied N-gram distribution.

Amirjalili et al. (2024) investigated dimensions of authorship in academic papers. The results indicate that human-authored texts are more complex, demonstrate more unique style and greater lexical diversity than AI-generated texts, whilst human-authored content exhibits unique strengths in the effective use of nuanced voice and rhetorical elements. The authors note that despite efforts to produce less generic style, AI-generated output remains formulaic, and whilst AI-generated text may appear human at first glance, it lacks the depth and authenticity of human-written work. This statement is supported by the study by Murcia Verdú et al. (2022), in which the authors examined source utilisation, narrative structure, style and word frequency to distinguish between AI and human-generated texts. The results demonstrate that AI-produced texts do not represent a qualitative contribution to (sports) journalism owing to their lack of analytical and interpretative character. Finally, attention is drawn to the empirical comparative analysis of lexicogrammatical features by Berber Sardinha (2024), who conducted his study along the five dimensions identified by Biber. This study revealed the following differences between AI-generated and human-written texts: (a) whilst human conversation is characterised by linguistic features capable of expressing speaker interactions and meaning negotiation, AI dialogue is characterised by a lack of engagement; (b) AI-generated text appears artificial, constructed and interrogative, lacking the hesitation, uncertainty and vagueness commonly found in spontaneous human conversation.

In our scoping review (Munn et al., 2018), we identified two studies (Nkhobo & Chaka, 2023; Zindela, 2023) which, albeit partially, demonstrated the benefits of AI-generated text—although of these authors, Zindela (2023) both employed a small sample and acknowledged in her study that the human-written texts were produced by students with poor or average language proficiency.

An empirical study by Desaire et al. (2023) established that text detectors designed to identify AI-written text (such as ZeroGPT, trained on 10 million documents and claiming 98 % accuracy in AI content detection) demonstrate insufficient effectiveness. As the authors identified error rates of 32 %, 42 % and 70 % across different topics and questions when testing detector effectiveness, they developed their own tool, which they report misclassified only one in 100 documents. Popkov and Barrett (2024) similarly identified problematic false positive and false negative rates when examining publicly available detectors, both free and subscription-based. Although Originality.AI has a specificity of .97, it shows more limited performance in detecting texts generated by the Claude software. The reliability of the ZeroGPT system is also questionable, as it falsely identified 62 % of human-written papers as partially generated by artificial intelligence.

Current research shows that AI-generated and human-written texts remain distinguishable, though with some limitations. However, multiple studies have raised concerns about the reliability of commercially available AI text detectors. Due to the high error rates and uncertainty associated with these publicly accessible detectors, we decided not to use them for authorship classification in our study. Additionally, our research context differed from previous studies since we lacked prior authorship information for the analysed texts. This meant that even established markers of AI-generated content, such as lexical diversity measures, could not provide reliable classification for our corpus comparison.

3. Method

3.1. Design of measuring instruments

The study employed a validated questionnaire comprising the following thematic blocks:

a) Background data: respondent's gender, parents' (or guardians') educational attainment, artificial intelligence use in learning (four items), and frequency of platform usage
b) Attitude scale: Schepman and Rodway's (2020) twenty-item validated questionnaire, The General Attitudes towards Artificial Intelligence Scale (GAAIS)
c) Essay: a non-handwritten student essay (homework) titled "My Views on AI-Generated Content"

Schools were instructed to assign the essay task to students two weeks before the questionnaire administration. Through the schools, we advised students to research the topic before writing their essays whilst emphasising that, regardless of their preparation, they should express only their own opinions and thoughts.

3.2. Sampling and data collection

To ensure high-quality data, we used a combination of convenience and random sampling. For convenience sampling, we included eight secondary schools from Budapest and Heves County (typically having multiple parallel classes per year), and an educational centre in Veszprém County that maintains four secondary institutions. Random selection was implemented within institutions: we randomly selected 1–2 classes from grades 9–12 from each participating secondary school. We supplemented the sampling by involving university students who work as teachers and distributed the questionnaire to their classes in their own schools, thus providing responses from additional institutions. Based on the participating institutions and individuals, the sample comprised approximately 30 classes (with an average of 22 students per class) from 20 secondary schools across different regions of the country. Data collection took place in April and May 2024. Students completed the online questionnaire, created on the Google Forms platform, in classroom settings during lessons, under teacher supervision, following a standardised procedure. The data collection method prevented multiple submissions, and questionnaire completion required answering all questions (except for uploading the essay). A total of 667 students completed the questionnaire.

3.3. Data cleansing

During data cleaning and coding, two reverse-coded items from the questionnaire were recoded. In the database, respondents were considered unreliable if at least one of the following conditions was met: responses showed a simple pattern (examining by question blocks, for example 1, 2, 3, 4, 5, 1, 2, 3, 4, 5); the variance of responses was remarkably low (examining by question blocks, for example 2, 2, 2, 2, 2, 2, 2, 2, 2, 2). Respondents were considered to be providing humorous or unreliable data if they, for instance, indicated living on a farm while reporting higher education level, both parents with incomplete primary education, and an age of 15 years. Respondents providing such and similarly inconsistent data (n = 10) were removed from the database, resulting in a sample size of 657. The respondents' ($N = 657$) minimum age was 14 years, maximum age was 21 years ($M = 16.23$, $SD = 1.186$). 36.5 % of respondents live in villages, 63.5 % in cities, 41.7 % are male, and 58.3 % are female.

3.4. Method of authorship attribution and ensuring the validity of sentiment analysis

Five independent coders evaluated the essays: three scholars with PhDs in educational sciences and two PhD students in educational sciences. The coders examined which essays might have been generated by artificial intelligence based on Berber Sardinha's (2024) criteria. Beyond these criteria, AI-generated content was also indicated if the essay:

a) contained no spelling, stylistic errors, or typos;

b) referenced that it could not have an opinion;
c) justified its opinion using bullet points;
d) deviated from the typical character of the age group studied, for example by arguing in a distinctly sophisticated manner or using technical terms;
e) presented pros and cons introduced by conjunctions in separate paragraphs, finally providing a summary with advice to balance opposing views;
f) when copied into a word processor, aligned paragraph endings to the right margin.

The coders evaluated the student essays according to three criteria: human-authored (code: 0) or software-generated (code: 1); meaningful for analysis (code: 1) or meaningless for analysis (code: 0); partial or complete plagiarism (noted as text annotation). Plagiarism was also checked using Turnitin software, independent of each coder's manual verification. Data cleaning occurred in three stages. In the first stage, coders received approximately 15 % ($n = 100$) of the total available corpus, performed test coding, and coding reliability was verified by calculating Krippendorff's alpha. However, the indicator proved unacceptably low ($\alpha < .5$). We identified one reason for this being that coders did not uniformly classify texts that were partly human-written, partly machine-generated, and partly or wholly plagiarised. In the first stage of data cleaning, besides standardising coding, coders discussed cases where two coders classified differently from the other three coders, but not due to different interpretations of the coding system. The questionable cases included essays which some coders believed were likely written with adult involvement or guidance due to the particular specificity or narrow technical nature of the topic selection. The coders discussed these questionable cases. We considered the first stage of data cleaning as coder training supported by cross-verification.

Following the coders' training, in the second stage of data cleaning, the coders recoded the entire available corpus ($n = 657$), and we calculated Krippendorff's alpha for coding reliability. Before calculating the indicator, we removed essays that contained no independently formulated sentences by students ($n_{\text{plagiarism}} = 16$). We retained 11 respondents in the database who submitted hybrid essays, but removed the plagiarised parts from their essays ($n_{1\text{-}3 \text{ sentences}} = 5$; $n_{4\text{-}10 \text{ sentences}} = 6$). Among these respondents were some ($n = 4$) who wrote their complete response partly plagiarised and partly using artificial intelligence. To ensure purely software-written texts were also available (plagiarised text was human-written in all cases), we removed the plagiarised parts from these corpora as well. The reliability indicator for the second round of coding approached an acceptable level ($\alpha = .701$, $CI_{95\ \%}$ [.681; .732]).

In the third stage of data cleaning, the coders discussed cases ($n = 43$) where two of the five coders evaluated essay authorship differently from the other three coders, and then classified the cases. In the final stage of data cleaning, we also removed respondents ($n = 91$) who neither wrote their essays using software nor plagiarised in whole or in part, but provided meaningless responses (empty fields, character strings, etc.). This includes translations of three foreign language texts, and manual review of several other items also resulted in their removal from the database for various reasons (for example, due to meaningless obscenity). As a result of data cleaning and human coding, we examined authorship for 548 essays. For sentiment analysis, we worked with a smaller sample as follows.

In the sentiment analysis, we excluded responses shorter than 50 characters, thus running the analysis on 450 observations. Prior to sentiment analysis, after removing control characters (line breaks, tabs, numbers), we lemmatised student responses using the HuSpaCy library. We addressed factors threatening the validity of sentiment analysis (Nandwani & Verma, 2021) as follows:

a) We considered negation structures. To account for negation structures, we created a list of all two- or three-element word pairs (bigrams, trigrams). We filtered these based on whether they contained at least one element from the sentiment dictionary and at least one element from a list of negation words. Examining the resulting list manually, we selected the relevant elements, which we replaced in the original texts by separating the word pair with an underscore, and then also inserted the word pair into the appropriate word list of the sentiment dictionary. The procedure resulted in three bigrams and zero trigrams entering the negative dictionary. Each of the three bigrams occurred once in the texts.
b) We considered ambiguity and irony. Through manual verification, we did not identify any ambiguity or irony in the text that would threaten the validity of sentiment analysis.
c) We considered response length. For proportional sentiment value, we counted positive and negative words in the corpora using the HunMinerR sentiment dictionary, subtracted the number of negative words from positive words (sentiment value), then calculated the total number of words for each response and divided the sentiment value by this.

3.5. Software used, statistical thresholds

Statistical analyses were conducted using SPSS 25 (regressions and reliability indices), JASP .18.1.0 (CFA), R and RStudio 4.3.0, and Python 3.12.3 (sentiment analysis). The absolute and relative fit indices of confirmatory factor analysis were evaluated based on the relevant literature (McNeish & Wolf, 2023) as follows: $\chi^2/df < 3$ (good); RMSEA <.08 (weak), RMSEA <.05 (good); RMSEA ULCI90 < .1 (good); CFI >.9 (acceptable), CFI >.95 (good); TLI >.9 (acceptable), TLI >.95 (good); IFI >.9 (acceptable), IFI >.95 (good); SRMR <.05 (good). The internal consistency indicator was examined using McDonald's ω based on Hayes and Coutts (2020) and Malkewitz et al. (2023). Similar to the reliability indicator calculated with Cronbach's alpha (Gliner et al., 2017, p. 186), we considered McDonald's ω values above .6 acceptable, above .7 adequate, and above .8 good. Krippendorff's alpha values above .8 were considered good. The parental educational attainment variable was created by averaging the two parents' scores measured on a ten-point ordinal scale, and was treated as a scale-level variable in the analysis. The unstructured data from the questionnaire, the student essays, were analysed using natural language processing (NLP) methods, specifically sentiment analysis (Rodríguez-Ibánez et al., 2023; Tan et al., 2023).

3.6. Characteristics of the scales

Attitudes towards artificial intelligence Schepman and Rodway's questionnaire, The General Attitudes towards Artificial Intelligence Scale (GAAIS), contains 20 items. The 20 items, rated on a five-point Likert scale, are organised into two subscales: positive attitudes (12 items) and negative attitudes (8 items) towards artificial intelligence. We examined this two-factor structure using confirmatory factor analysis (CFA), as the original questionnaire was validated on a different population. Therefore, we verified whether the factor structure was valid for the sample in the present research (reliability indices alone are insufficient for this purpose, as they only measure internal consistency and do not provide information about the adequacy of the factor structure). The fit indices were adequate ($\chi^2 = 733.713$, $df = 169$, $\chi^2/df = 4.341$, $p < .001$, RMSEA = .071, SRMR = .062, TLI = .839, CFI = .857, IFI = .857). As we aimed to create composite variables that explain the maximum variance from the original variables, we used principal component analysis (PCA) to identify latent variables that could be included in further analyses. Based on the Kaiser-Meyer-Olkin value, the results of the principal component analysis were interpretable (KMO = .883), and Bartlett's test showed significant correlations between items [$\chi^2(df = 190, N = 657) = 4073.412$, $p < .001$]: based on the correlation matrix $.333 \leq r \leq .671$. The determinant of the correlation matrix (D = .002) significantly exceeded the criterion value of .00001, thus problematic multicollinearity could be ruled out. Based on Kaiser's criterion, we retained principal components with eigenvalues greater than 1. The

initial data reduction procedure identified four principal components with eigenvalues above 1 from the original 20 items, which together explained 52.556 % of the total variance of the items [Table 1].

As the eigenvalues of the third and fourth principal components barely exceeded 1, contributed minimally to the explained variance, and the fourth principal component relied on only two items, we reran the analysis without the seven items from these components. The Kaiser-Meyer-Olkin value indicated interpretable results for the second principal component analysis (KMO = .851), and Bartlett's test showed significant correlations between items [$\chi^2(df = 78, N = 657) = 2303.115, p < .001$]. The determinant of the correlation matrix (D = .002) substantially exceeded the criterion value of .00001, thus problematic multicollinearity could be ruled out. Following dimension reduction, we applied orthogonal varimax rotation and extracted subscale scores using the Anderson-Rubin method to ensure orthogonality. The second data reduction procedure identified two principal components with eigenvalues above 1 from the remaining 13 items, collectively explaining 46.842 % of the total item variance [Table 2].

The first dimension, 'negative AI attitudes', contains six items and has good scale reliability ($\omega = .815$). The second dimension, 'positive AI attitudes', contains seven items and has adequate scale reliability ($\omega = .759$).

Artificial intelligence in learning Respondents rated four items on a four-point Likert scale regarding the role of artificial intelligence in their learning. We examined the structure of responses using principal component analysis (PCA). Bartlett's test showed significant correlations between items [$\chi^2(df = 6, n = 657) = 299.517, p < .001$], and the Kaiser-Meyer-Olkin value indicated interpretable results for the principal component analysis (KMO = .713). The analysis identified a single principal component explaining 48.050 % of the total item variance. The dimension labelled "Artificial Intelligence in Learning" consists of four items, with the scale showing acceptable reliability ($\omega = .637$) [Table 3].

Frequency of platform use Respondents indicated the daily time spent using three known AI platforms on a seven-point Likert scale. Principal component analysis (PCA) was used to examine the structure of the responses. Based on Bartlett's test, there were significant correlations between the items [$\chi2(df = 3, N = 657) = 89.562, p < .001$], but based on the Kaiser-Mayer-Olkin value, the results of the principal component analysis were only tentatively interpretable (KMO = .543). The single principal component created explained 46.816 % of the total variance of the items. The dimension 'frequency of platform use' contains three items and the scale reliability is not adequate ($\omega = .453$) [Table 4].

The relatively weak dimensionality reduction indicators are technically understandable: users of one platform are likely to use others less frequently or not at all. However, when considering the scale's inclusion in further analyses, we examined not only the low KMO and ω values but also the Bartlett's test result (adequate), communalities (adequate in two cases, low in one) and explained variance (acceptable). Consequently, we included the platform usage frequency dimension in the multiple linear regression model but not in the logistic regression model. For the latter, we utilised the three ordinal scales that constitute the dimension.

Table 1
Principal components of the 20-item GAAIS questionnaire.

Items	1	2	3	4
I think Artificial Intelligence is dangerous	.810			
Artificial Intelligence might take control of people	.777			
I find Artificial Intelligence sinister	.762			
I shiver with discomfort when I think about future uses of Artificial Intelligence	.687			
Artificial Intelligence is used to spy on people	.539			
People like me will suffer if Artificial Intelligence is used more and more	.528			
For routine transactions, I would rather interact with an artificially intelligent system than with a human		.696		
I would like to use Artificial Intelligence in my own job		.646		
An artificially intelligent agent would be better than an employee in many routine jobs		.603		
I am interested in using artificially intelligent systems in my daily life		.590		
Artificially intelligent systems can perform better than humans		.568		
Artificial Intelligence can provide new economic opportunities for this country		.546		
Artificially intelligent systems can help people feel happier		.472		
I am impressed by what Artificial Intelligence can do			.791	
Artificial Intelligence is exciting			.757	
There are many beneficial applications of Artificial Intelligence			.733	
Artificial Intelligence can have positive impacts on people's wellbeing		.407	.501	
Much of society will benefit from a future full of Artificial Intelligence		.435	.483	
I think artificially intelligent systems make many errors				.738
Organisations use Artificial Intelligence unethically				.633

The columns 1, 2, 3, 4 of the table show the factor loadings corresponding to each principal component.

Table 2
Principal components of the 13-item GAAIS questionnaire.

Items	1	2
I think Artificial Intelligence is dangerous	.829	
I find Artificial Intelligence sinister	.790	
Artificial Intelligence might take control of people	.728	
I shiver with discomfort when I think about future uses of Artificial Intelligence	.703	
Artificial Intelligence is used to spy on people	.597	
People like me will suffer if Artificial Intelligence is used more and more	.576	
I am interested in using artificially intelligent systems in my daily life		.677
I would like to use Artificial Intelligence in my own job		.676
Artificial Intelligence can provide new economic opportunities for this country		.632
An artificially intelligent agent would be better than an employee in many routine jobs		.632
For routine transactions, I would rather interact with an artificially intelligent system than with a human		.611
Artificially intelligent systems can help people feel happier		.587
Artificially intelligent systems can perform better than humans		.577

Columns 1 and 2 of the table show the factor loadings corresponding to each principal component.

4. Results

4.1. Authorship of essays (human or software)

After data cleansing, the reliability of the coding is good ($\alpha = .829$, $CI_{95\ \%}$ [.808; .850]), with 463 sentences written by humans and 85 sentences written by AI ($N_{total} = 548$) in the analysable database. In our

Table 3
Principal component of the questionnaire on the use of AI in learning.

Items	1
How much would you mind if teachers also used AI in teaching or assessment?	.717
How effective do you think artificial intelligence is at learning?	.712
The use of artificial intelligence will be essential in my studies in the near future	.689
The use of artificial intelligence distorts or falsifies the results and conduct of studies	.653

Column 1 of the table shows the factor loadings corresponding to the principal component.

Table 4
Principal component of frequency of platform use.

Items	1
How much time per day do you actively use it? – Google Gemini/Bard	.770
How much time per day do you actively use it? – Microsoft Bing	.765
How much time per day do you actively use it? – ChatGPT	.476

Column 1 of the table shows the factor loadings corresponding to the principal component.

study, approximately 16 % of respondents wrote their sentences using AI.

4.2. Factors determining the sentiment, authorship (human or software) and use of AI in learning

We examined factors determining the use of artificial intelligence for learning purposes and attitudes towards AI-generated content using two linear regression models [Table 5]. Using logistic regression, we analysed the extent to which gender, parents' average educational level, platform usage frequency (separately for three variables), negative and positive attitudes towards artificial intelligence, and the use of artificial intelligence for learning purposes predict whether students will use artificial intelligence for their text-based homework assignments [Table 6]. For the logistic regression, we employed a hierarchical approach, which enabled a deeper understanding of effect mechanisms. This method offered several advantages: firstly, we were able to identify suppressor effects, namely cases where a variable only became significant or lost its significance after controlling for other variables. Secondly, we could examine the robustness of effects, as when a variable's effect (such as gender) remained stable across all models, it provided stronger evidence for the existence of that effect. Moreover, the hierarchical method demonstrated how the coefficients of previously included variables changed with the introduction of new variables. Following the structure of our theoretical model, we first incorporated basic demographic variables (gender, parents' educational attainment) and then built additional layers upon these (platform usage, attitudes).

We checked the assumptions of multiple regression as follows. For screening multidimensional outliers in both linear and logistic regression analyses, we calculated Cook's distances. We examined the data using a criterion level of 1. According to the results, there were no multidimensional outliers in the data (maximum value: .04). Based on tests suitable for checking the collinearity assumption, neither collinearity nor multicollinearity was detected between the variables. The high independence of predictors in both multiple linear models allowed for a clear interpretation of their effects; therefore, variables were entered into the regressions using the ENTER method (for logistic regression in blocks, see Model 1, 2, 3, 4, 5). Based on the scatterplot showing standardised predicted values and standardised residual errors, the assumptions of homoscedasticity and linearity were met for both linear analyses. Based on the histograms, the assumption of normality of standardised residual errors was met. The assumption of independence of residual errors was also met (Durbin-Watson statistic in the two models was 1.835 and 1.940).

Table 5
Factors determining the use of artificial intelligence in learning and the sentiment of students' essays.

Dependent variable	Predictors	β	t	p	Tolerance
First model	Intercept		.89	.373	
	Age	−.05	−1.61	.108	.982
	Average educational attainment of parents	−.02	−.63	.532	.970
	Gender	.11	3.24	.001	.892
	Negative artificial intelligence attitudes	−.33	−9.74	<.001	.909
	Positive artificial intelligence attitudes	.50	15.65	<.001	.982
Second model	Intercept		1.93	.054	
	Average educational attainment of parents	.02	.34	.733	.992
	Gender	.02	.50	.620	.991
	Artificial intelligence in learning	.100	1.95	.052	.885
	Frequency of platform use	.00	−.06	.955	.880

Dependent variable of the first model: artificial intelligence in learning, N = 657; dependent variable of the second model: sentiment of the student essay "My opinion on artificial intelligence-generated content", N = 450.

The results show that the first model is significant (F [5, 651] = 66.625; $p < .001$; $R^2 = .339$; adj. $R^2 = .333$), the explained variance is adequate (34 %), and based on tolerance values, the overlap between predictors is low. The second model is not significant (F [4, 445] = 1.104; $p = .354$; $R^2 = .010$; adj. $R^2 = .001$), the explained variance is low (1 %), and based on the high tolerance values, the overlap between predictors is also low [Table 5].

Based on the results, positive attitudes towards artificial intelligence have a significantly strong positive effect, while negative attitudes towards artificial intelligence have a significantly strong negative effect on the dependent variable, indicating that students' attitudes towards artificial intelligence significantly determine its use in learning. This result was expected, and examining it strengthens the validity of unexpected findings: the use of artificial intelligence for learning purposes is not significantly influenced by age or parents' average educational level. Gender is the only demographic variable that has a significant effect on the dependent variable: women have more positive views about using artificial intelligence in learning.

The second model is not significant. This can be explained by the null hypothesis being true, meaning there is indeed no effect, and the model does not include the variables that explain students' attitudes towards artificial intelligence, or insufficient evidence was gathered in the present study to reject the null hypothesis (Visentin et al., 2020).

The results of the logistic regression are shown in Table 6 [Table 6].

Based on the first significant model, parents' average educational level does not influence authorship. However, the gender effect is significant: women (coding 1: male, 2: female) are less likely to use artificial intelligence for completing their homework, with gender approximately halving the likelihood of use (Exp(B) = .42).

The second model is significant, but the block itself is not, meaning the three platform usage variables did not significantly increase the explained variance. Nevertheless, including the block was justified: in the second model, although the gender effect weakens slightly, it remains strongly significant, while ChatGPT usage also emerges as a significant predictor. More frequent ChatGPT usage increases the likelihood of having homework written by artificial intelligence (by approximately one-third). The effects of the other two platforms are not significant, which may be explained by the correlation between gender and ChatGPT usage: boys use ChatGPT more frequently.

In the third significant model, the effect of using artificial intelligence for learning purposes is not significant and does not increase the explanatory power of the previous model, thus based on the results, the use of artificial intelligence in learning does not influence homework authorship. ChatGPT usage loses its significant explanatory power, presumably because the explanatory power is distributed between the learning-related use and ChatGPT usage variables.

The fourth significant model shows a similar pattern regarding positive attitudes towards artificial intelligence: notably, positive attitudes do not increase the likelihood of students having their papers written by artificial intelligence.

In the fifth significant model, however, the significant effect of negative attitudes towards artificial intelligence emerges, and the change compared to the previous block is also significant. The gender effect maintains its significance, and the negative attitudes scale proves

Table 6
Factors determining authorship.

Model	Predictors	B	S.E.	Wald	p	Exp(B)	95 % C.I. for EXP(B)	
Model 1	Gender	−.86	.24	12.64	.000	.42	.26	.68
	Average educational attainment of parents	.07	.07	1.06	.304	1.07	.94	1.23
	Constant	−.91	.65	1.92	.165	.40		
Model 2	Gender	−.77	.25	9.62	.002	.46	.28	.75
	Average educational attainment of parents	.08	.07	1.47	.226	1.09	.95	1.25
	ChatGPT daily use	.24	.12	4.15	.042	1.27	1.01	1.61
	Google Gemini/Bard daily use	−.06	.10	.33	.564	.95	.78	1.14
	Microsoft Bing daily use	.06	.18	.11	.742	1.06	.75	1.51
	Constant	−1.51	.74	4.14	.042	.22		
Model 3	Gender	−.77	.25	9.44	.002	.46	.28	.76
	Average educational attainment of parents	.09	.07	1.53	.216	1.09	.95	1.25
	ChatGPT daily use	.21	.13	2.59	.107	1.23	.96	1.58
	Google Gemini/Bard daily use	−.07	.10	.44	.507	.94	.77	1.14
	Microsoft Bing daily use	.06	.18	.09	.759	1.06	.74	1.50
	Artificial intelligence in learning	.100	.13	.56	.453	1.10	.85	1.42
	Constant	−1.46	.75	3.79	.052	.23		
Model 4	Gender	−.77	.25	9.42	.002	.46	.28	.76
	Average educational attainment of parents	.09	.07	1.53	.217	1.09	.95	1.25
	ChatGPT daily use	.21	.13	2.60	.107	1.23	.96	1.59
	Google Gemini/Bard daily use	−.07	.100	.44	.510	.94	.77	1.14
	Microsoft Bing daily use	.06	.18	.10	.755	1.06	.74	1.50
	Artificial intelligence in learning	.11	.14	.54	.463	1.11	.84	1.47
	Positive artificial intelligence attitudes	−.02	.13	.02	.904	.98	.76	1.28
	Constant	−1.46	.75	3.79	.051	.23		
Model 5	Gender	−.62	.26	5.56	.018	.54	.32	.90
	Average educational attainment of parents	.10	.07	1.89	.169	1.10	.96	1.26
	ChatGPT daily use	.20	.13	2.45	.118	1.23	.95	1.58
	Google Gemini/Bard daily use	−.06	.10	.32	.575	.95	.78	1.15
	Microsoft Bing daily use	.07	.18	.14	.706	1.07	.76	1.52
	Artificial intelligence in learning	−.01	.15	.00	.970	.99	.74	1.34
	Positive artificial intelligence attitudes	.05	.14	.11	.742	1.05	.80	1.36
	Negative artificial intelligence attitudes	−.28	.14	4.32	.038	.76	.58	.98
	Constant	−1.82	.78	5.49	.019	.16		

Dependent variable in all models: the authorship (human or software) of the essays. Model 1. $\chi^2(2) = 14.82$, $p = .001$, Cox & Snell $R^2 = .03$, Nagelkerke $R^2 = .05$.; Model 2. $\chi^2(5) = 18.98$, $p = .002$, Cox&Snell $R^2 = .03$, Nagelkerke $R^2 = .06$, $\Delta\chi^2(3) = 4.16$, $p = .245$; Model 3. $\chi^2(6) = 19.54$, $p = .003$, Cox&Snell $R^2 = .04$, Nagelkerke $R^2 = .06$., $\Delta\chi^2(1) = .56$, $p = .452$; Model 4. $\chi^2(7) = 19.56$, $p = .007$, Cox&Snell $R^2 = .04$; Nagelkerke $R^2 = .06$, $\Delta\chi^2(1) = .16$, $p = .904$; Model 5. $\chi^2(8) = 23.91$, $p = .002$, Cox&Snell $R^2 = .04$, Nagelkerke $R^2 = .07$, $\Delta\chi^2(1) = 4.36$, $p = .037$.

to be a stronger predictor than the positive attitudes scale. The results indicate that women and those who are more critical of artificial intelligence are less likely to have their homework written using artificial intelligence.

The gradual introduction of blocks in the hierarchical regression analysis revealed important relationships. The first block, containing demographic variables, demonstrated the strong effect of gender, which remained stable throughout subsequent models. The second block, comprising platform usage variables, although it did not significantly increase the explained variance on its own, revealed the relationship between ChatGPT usage and gender. The AI learning usage variable introduced in the third block highlighted an important suppressor effect: ChatGPT usage lost its significant explanatory power as it was shared with learning-related usage. The introduction of attitude variables in the fourth and fifth blocks revealed an interesting pattern: whilst positive attitudes did not significantly influence homework authorship, negative attitudes did.

Based on the above results, as the final step of the analysis, we examined which specific types of negative attitudes towards artificial intelligence correlate with authorship. For this, we analysed the correlations between the original 20 items of the dimension and authorship [Table 7].

Based on the results, authorship is only significantly determined by items 8 (I find Artificial Intelligence sinister), 10 (I think Artificial Intelligence is dangerous) and 15 (I shiver with discomfort when I think about future uses of Artificial Intelligence). Comparing these three statements with the remaining five items of the negative attitude scale (eight items in total), the statements refer more to general fears, i.e. they do not contain as much specificity as, for example, items 6 (I think artificially intelligent systems make many errors) or 20 (Artificial Intelligence is used to spy on people).

5. Discussion and implications

In our study, 16 % of students wrote their text-based homework assignments using artificial intelligence (answer to Q1), which is lower than the self-reported 33 % among primary school students (Li et al., 2024). Beyond natural fluctuation and age differences between the two study samples, this difference might also be attributed to people clearly distinguishing between tasks where they accept the use of artificial intelligence and those where they accept it less (Bellaiche et al., 2023; Ofosu-Ampong, 2024; Tang et al., 2023). While such limited empirical evidence is certainly insufficient for drawing well-founded conclusions, it is probable that students use software solutions more frequently for homework requiring longer text composition than for calculation or editing tasks. We note that essays written at home with the assistance of artificial intelligence may vary in quality regarding the detectability of authorship (Herbold et al., 2023), meaning that by adding punctuation or typing errors, software-generated text may more likely appear as human-composed writing to evaluators (Ibrahim et al., 2023). This must be considered in teachers' daily assessment practices, when working with software detectors, in coding, and generally in future research.

In our study with a medium-sized sample, we could not identify factors determining the sentiment of compositions (answer to Q2). This might be because student sentiments in the secondary school age group are diffuse, random, and unfounded, meaning they currently lack a latent structure that would stably connect to the examined demographic and other variables. This notion is also supported by our finding that negative attitudes towards artificial intelligence are general and barely connected to specifics. The result suggests that the examined sample

Table 7
The General Attitudes towards Artificial Intelligence Scale (GAAIS) in relation to authorship.

	Authorship	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Authorship																					
1																					
2		.28																			
3																					
4		.25	.24																		
5		.11	.22		.23																
6		−.9		.27		−.16															
7		.31	.31		.35	.27	−.15														
8	−.12	−.24	−.12	.31	−.17		.29	−.27													
9		−.19		.12	−.14			−.13	.53												
10	−.17	−.19		.29	−.2		.26	−.24	.66	.57											
11		.28	.31	−.8	.47	.3	−.8	.38	−.26	−.23	−.26										
12		.28	.28		.31	.52	−.8	.44	−.22	−.2	−.17	.4									
13		.42	.28		.28	.27	−.1	.32	−.24	−.17	−.25	.32	.29								
14		.24	.32		.26	.46	−.8	.31	−.13	−.13	−.14	.41	.52	.37							
15	−.13	−.14	−.8	.19	−.15	−.9	.17	−.19	.5	.39	.52	−.23	−.2	−.22	−.18						
16		.21	.23		.17	.25	−.11	.23				.16	.23	.3	.23						
17		.22	.34		.3	.29		.39	−.25	−.18	−.2	.41	.37	.31	.39	−.24	.26				
18		.32	.29	−.13	.35	.22	−.17	.54	−.35	−.14	−.25	.36	.29	.29	.27	−.23	.26	.4			
19		−.9	−.8	.21	−.1	−.8	.15	−.19	.36	.25	.38	−.14	−.9	−.15	−.12	.41		−.21	−.27		
20				.24		.1	.2	−.9	.36	.3	.4	−.8				.28			−.9	.21	

Spearman *r*. $N = 548$ (authorship), $N = 657$ (attitudes). Only correlations significant at the $p < .05$ level are shown. 1. For routine transactions, I would rather interact with an artificially intelligent system than with a human; 2. Artificial Intelligence can provide new economic opportunities for this country; 3. Organisations use Artificial Intelligence unethically; 4. Artificially intelligent systems can help people feel happier; 5. I am impressed by what Artificial Intelligence can do; 6. I think artificially intelligent systems make many errors; 7. I am interested in using artificially intelligent systems in my daily life; 8. I find Artificial Intelligence sinister; 9. Artificial Intelligence might take control of people; 10. I think Artificial Intelligence is dangerous; 11. Artificial Intelligence can have positive impacts on people's wellbeing; 12. Artificial Intelligence is exciting; 13. An artificially intelligent agent would be better than an employee in many routine jobs; 14. There are many beneficial applications of Artificial Intelligence; 15. I shiver with discomfort when I think about future uses of Artificial Intelligence; 16. Artificially intelligent systems can perform better than humans; 17. Much of society will benefit from a future full of Artificial Intelligence; 18. I would like to use Artificial Intelligence in my own job; 19. People like me will suffer if Artificial Intelligence is used more and more; 20. Artificial Intelligence is used to spy on people.

does not yet possess well-founded knowledge about artificial intelligence. As attitudes towards artificial intelligence are also influenced by knowledge of it (Kaya et al., 2024), our recommendation for attitude formation remains curriculum development and organisation of training programmes. It is important to note that users inherently judge artificial intelligence use more positively for working with large databases and easily automated tasks, whilst more negatively for tasks involving understanding, empathy, and artistic activities (Schepman & Rodway, 2020).

The authorship of compositions assigned as homework is determined by gender and attitudes towards artificial intelligence. Based on the results, women and those who are more critical of artificial intelligence are less likely to write their homework using artificial intelligence (answer to Q3). Such an effect of attitudes is logically expected and not surprising based on empirical precedents (Kim & Lee, 2024). The gender effect partially contradicts the literature (Kim & Lee, 2024), which might be explained by Pellas (2023) demonstrating positive attitudes in 73 % of university students and the limited role of demographic factors determining positive attitudes. Therefore, future studies are needed to definitively demonstrate the role of gender effect in determining attitudes among secondary school students. For this, we recommend using newer measurement tools (Stein et al., 2024).

In our study, the use of artificial intelligence in learning was not influenced by age or parents' average educational level. However, women held more positive views regarding the application of artificial intelligence in learning (answer to Q4). This result may reflect general attitudinal differences between genders, suggesting that women are more open to new learning methods. The absence of effects from age and parental educational level suggests that attitudes towards artificial intelligence are not tied to generational differences and are independent of family background. Our results thus confirm the assumption that artificial intelligence has a democratising effect, meaning its use contributes to equalising differences between students from various backgrounds.

5.1. Pedagogical considerations

Artificial intelligence simultaneously projects the image of both human replacement and human development. It is conceivable that whilst the developers' original intention was for artificial intelligence to function as an assistant developing individuals, in a negative scenario this very feature could lead to the replacement and regression of individuals. Another unintended consequence could be that artificial intelligence will be capable of hindering the development of teachers' digital competencies (this could occur, for instance, if teachers only control systems by voice and no longer understand their operation). Artificial intelligence systems will soon influence every aspect of learning: personal skills, knowledge acquisition, learning ability and career building. From a pedagogical perspective, it is significant that individuals' current use of technology is often intuitive rather than learned. In our view, regarding the use of artificial intelligence, we can distinguish four groups now and in the near future:

1 One part of humanity supports the robust use of digital technology and especially artificial intelligence. They use it to become even more intelligent through personalised development and to enhance their competencies. They manage to live in a kind of symbiosis with the machine, as described by Licklider. According to Licklider's pioneering concept, the close partnership between humans and electronic computers facilitates thinking and enables cooperative decision-making. Through this symbiosis, various operations can be performed much more efficiently than by humans alone, and this group of humanity embarks on a rapid development trajectory.
2. Another part of humanity uses artificial intelligence technology, but primarily to replace their own work (in a pessimistic reading, their own existence) out of convenience, drift, or other reasons, whilst not even being aware of it. Passive use may lead to uninformed usage, cognitive dependency (cf. Suriano et al., 2025), and a state of 'drifting'. The 'drifters' schedule is determined by the arrival of social media content, emails, chat messages, and the compulsion to respond. Those who use artificil intelligence to perform their routine tasks are doing nothing different than when they use a machine to wash, travel, or grind their coffee. However, those who outsource tasks that would ensure their cognitive development to artificial intelligence do not develop. They act similarly to parents or teachers who allow children to use technology inappropriately from a very young age, resulting in the non-development of age-appropriate skills.
3. The third part of humanity opposes the use of technology. They are few in number, as the subsystems of our society are increasingly permeated by digitalisation and networks, and will be increasingly permeated by artificial intelligence. Individuals will find it increasingly difficult to conduct their shopping, public affairs, or social interactions without using technology. All this applies to education as well.
4. The fourth group simply has no access to artificial intelligence. The part of humanity that does not use technology thus remains on the 20th century's development curve and does not develop at the same rate as the first three groups.

The differences in education are already visible. Drifters are already having their essays written with the help of ChatGPT, submitting them at school without any processing or criticism, whilst in the case of educators, automatic task distribution and machine evaluation will result in their complete detachment from students. In contrast, the 'governing educators' belonging to the first group train students to properly parametrise and use artificial intelligence during learning, as they will need to proceed similarly in the world of work. The contradictions between governors and drifters can only be bridged if the world of education finds the balance between technology and nature. We must therefore seize every opportunity to raise awareness and teach that humanity must use technology responsibly, purposefully, and consciously.

In the digital competency framework, the conscious use of artificial intelligence is one of the most important competencies needed in the 21st century. In our view, the question of conscious use of artificial intelligence can be divided into three main elements: 1. Understanding the basic concepts and operating principles of artificial intelligence: students should be able to understand machine learning, natural language processing, and various applications of artificial intelligence. 2. Knowledge of ethical and legal aspects of artificial intelligence: students should be able to assess the potential ethical and legal risks of artificial intelligence and take these into account when designing and using systems. 3. Critical evaluation of artificial intelligence-based systems, so that students can recognise potential errors and limitations of the systems.

It is of paramount importance in teacher training to identify the benefits associated with artificial intelligence (cf. Fu & Weng, 2024). These include that solutions delivered by artificial intelligence are always fast and automate processes. It provides complete textual responses that are easily useable and handles routine communication tasks with proper parametrisation. It argues and teaches argumentation to those who want to understand its operation. It is both an advantage and a challenge that artificial intelligence generates non-existent text, therefore according to plagiarism detection systems, it does not plagiarise. We would not exercise due diligence if we did not discuss with teacher training participants the greatest challenge currently experienced: artificial intelligence can be easily and quickly used for cheating. It can easily solve homework, and the solution can be submitted as one's own. However, cheating is self-deception, a key element of drifting. Such replacement of human intellect endangers existing pedagogical practices, therefore we must redefine concepts such as homework. This, however, is also an advantage, as it will bring cultivation and thinking to the forefront, as opposed to mechanical collection and recitation of

knowledge or routine exploration of connections.

5.2. Practical implications

The practical recommendations of our research are timely and require swift action from multiple perspectives. The revealed 16 % artificial intelligence usage rate necessitates intervention in rethinking school assessment practices. Changes are also timely needed in the field of teacher training. We observe that modification of training programmes is necessary, with particular attention to teaching the ethical use of artificial intelligence. In the area of digital competence development, we identify three areas: a) understanding the basic concepts and operating principles of artificial intelligence; b) knowledge of the ethical and legal aspects of artificial intelligence; c) critical evaluation of artificial intelligence-based systems. These areas require immediate development for effective and responsible use. The need for institutional-level responses is equally urgent. Our research indicates that educational institutions need to develop guidelines for the use of artificial intelligence and implement appropriate monitoring mechanisms.

6. Conclusion

Our study had two objectives: firstly, to determine through human coding the authorship of student essays submitted as homework (human or software), and secondly, to examine the sentiment of these essays and the factors determining the use of artificial intelligence in learning. The main research problem was that whilst practical experiences and news articles suggest that a significant proportion of students use artificial intelligence for text creation tasks, there is scarcely any empirical, reliable data available on this. To address this question, we examined students from thirty classes across twenty Hungarian secondary schools ($N = 548$) using a validated questionnaire and essays analysed through human coding. Our results show that 16 % of secondary school students use artificial intelligence for their written homework assignments, which is lower than the previously measured 33 % among primary school students (based on self-reporting). The use of artificial intelligence in learning is not influenced by age or parents' educational attainment; however, whilst women have more positive views about it, they are less likely to use it for completing homework. The sentiment analysis did not reveal any stable patterns, suggesting that the studied age group does not yet possess well-founded knowledge and crystallised opinions about artificial intelligence. The negative attitudes towards artificial intelligence are general and lack specificity, which also indicates a lack of knowledge. Our results confirm the democratising effect of artificial intelligence, as its use is independent of family background and age. Based on the identified correlations, we recommend curriculum development and the organisation of further training programmes, with particular emphasis on teaching the ethical and effective use of artificial intelligence.

6.1. Limitations and future research

Our research has several limitations. Firstly, whilst the reliability of human coding was adequate ($\alpha = .829$), this method of determining authorship cannot be considered perfect. As indicated in the literature, texts written by artificial intelligence can be of varying quality in terms of detectability: by adding punctuation marks or typing errors, a software-generated text may be more likely to appear as human-composed writing. A second limitation is that in our study, we were unable to identify factors determining the sentiment of the compositions in our medium-sized sample. Whilst this could be explained by the fact that student sentiments are still diffuse and random in the secondary school age group, the size and composition of the sample may have also influenced the results. Regarding sampling, although we strived for representativeness, the sample was partly based on availability. Although we collected data from twenty secondary schools across different regions of the country, the sample cannot be considered fully representative of the Hungarian secondary school population.

Several suggestions can be made for future research. As emphasised in the literature, a more thorough analysis of the ethical aspects of artificial intelligence in various educational contexts is needed (Ifenthaler et al., 2024; Rojas & Chiappe, 2024). It would be particularly important to further explore the relationships between student attitudes and artificial intelligence usage (Tsekhmister et al., 2024), as well as to examine the application of the technology at other educational levels (García-López et al., 2025). The paradox revealed in gender differences –that women view the use of artificial intelligence in learning more positively yet are less likely to use it – requires further investigation. It would be worth examining this relationship with a larger sample and different methodological approaches, with particular attention to the underlying causes. Longitudinal studies are needed to understand the long-term effects of artificial intelligence use, particularly regarding the development of critical thinking and learner autonomy. This aligns with Murtiningsih et al.'s (2024) suggestion regarding the examination of long-term effects.

Finally, further research is needed into the dynamics and characteristics of the four user groups we identified (governors, drifters, opponents, outsiders), as well as the conditions for transition between groups. A deeper understanding of the differences between groups could help in developing more targeted educational programmes.

CRediT authorship contribution statement

Mátyás Turós: Conceptualization, Methodology, Software, Validation, Formal analysis, Data Curation, Writing – original draft, Writing – review & editing, Visualization. **Róbert Nagy:** Investigation, Writing – original draft. **Zoltán Szűts:** Conceptualization, Investigation, Resources, Supervision, Project administration, Funding acquisition.

Statements and Declarations

The study was approved by an ethical committee with ID: RK/1560/2024. Informed consent was obtained from all participants, and their privacy rights were strictly observed.

Statement on open data

The data were placed in a public repository: https://doi.org/10.7910/DVN/MJRS2D.

Declaration of AI assisted writing

No generative artificial intelligence (AI) has been used in this scientific work.

Funding statement

The research was funded by the Hungarian National Media and Infocommunications Authority (Nemzeti Média -és Hírközlési Hatóság; https://nmhh.hu). Grant number: 002/2024.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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